

features in ROCm 7 is support for FP4 and FP6, two ultra-low-precision data formats designed for high-throughput inference. These formats are particularly relevant to AMD's new Instinct MI350X and MI355X accelerators, which are built on the CDNA 4 architecture and have native support for FP4 and FP6 matrix operations.

For comparison, NVIDIA's TensorRT-LLM and Hopper architecture lean heavily into FP8 and INT4, so AMD's adoption of FP4/FP6 keeps them competitive on the efficiency front. These formats reduce memory bandwidth and power consumption while maintaining accuracy for AI inference tasks.

Distributed Inference and Training Support: Large model deployment isn't just about one GPU anymore. ROCm 7 brings native support for distributed inference and training, with full integration into frameworks like vLLM, SGLang, and Llm-d. These are the backbones for production-grade LLM serving pipelines, enabling efficient attention caching, dynamic batching, and multi-node scaling. This is especially relevant for developers deploying LLM chatbots, search engines, and recommendation systems, where latency and throughput are critical.

Enterprise AI Solutions and MLOps: ROCm 7 is also introducing an Enterprise AI layer that's targeted at production deployment. This includes a ROCm MLOps stack with model fine-tuning, secure scaling, logging, and integration with tools like Kubernetes, MLFlow, and Prometheus. In other words, ROCm is positioning itself as a viable AI operations backbone. There's also built-in support for AI workload security policies, role-based access control, and encrypted model storage which are all necessary features for deployment in regulated industries.

Hardware and OS Support

Radeon and Ryzen AI Support: One of the most noticeable changes with ROCm 7 is its expanded hardware compatibility. Developers can now tap

The graphic is divided into two main sections. The top section, titled 'In-Box Linux Support' and 'Full Windows Support', lists various operating systems and frameworks. The bottom section, titled 'The ultimate Client AI solutions for every need', showcases three laptop configurations: AMD Ryzen AI 300 (Up to 24B parameters), AMD Ryzen AI Max (Up to 70B parameters), and Threadripper + Radeon AI (Up to 128B parameters).

In-Box Linux Support	Full Windows Support
Red Hat EPEL (2H 2025)	PyTorch (Preview Q3 2025)
Ubuntu (2H 2025)	ONNX-EP (Preview July 2025)
OpenSUSE (NEW)	HIP SDK
Fedora	Linux in Windows WSL

The ultimate Client AI solutions for every need		
AMD Ryzen AI 300 Up to 24B parameters	AMD Ryzen AI Max Up to 70B parameters	Threadripper + Radeon AI Up to 128B parameters

into ROCm on select Radeon GPUs and Ryzen AI processors. While performance on Radeon cards won't match Instinct-class accelerators, it gives students, hobbyists, and prosumers a more affordable way to experiment with ROCm and AI workloads.

This shift could be key to driving grassroots developer adoption, an area where NVIDIA has historically held a significant edge with CUDA.

Arguably the most overdue feature, ROCm 7 supports Windows. Official support includes native Windows and WSL2 environments, meaning data scientists who primarily use Visual Studio Code can join the ROCm ecosystem.

This is a strategic move. With so many AI developers on Windows (especially in enterprise and education), AMD is opening the gates to a user base that relied on dual-booting Linux or using Docker containers.

Experience and Ecosystem

Bi-Weekly Updates and Day-Zero Fixes: AMD is adopting a modern software delivery with ROCm 7. Instead of large, infrequent releases, it will push bi-weekly updates with day-zero bug fixes and patch notes. This brings ROCm closer to modern expectations and matches the velocity of open-source machine learning tooling.

This is AMD's version of NVIDIA's DGX Cloud but with full ROCm integration. Developers can test LLMs, tune models, and benchmark, all without needing to invest in physical hardware.

CUDA Porting Made Easier: ROCm 7 continues to improve HIPIFY, the tool AMD provides to convert CUDA code into ROCm-compatible HIP code. [d](https://tinyurl.com/roc7md)

Read more on Digit.in:
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